SOFTWARE REQUIREMENTS SPECIFICATION / STUDENT EDITION

PhishShift Message Risk Scanner

Project Code: ML-07 | Focus: Cybersecurity NLP + Robustness

Level: Intermediate | Duration: 4 weeks | Submission: Working prototype + evidence-based report

01 | Problem Statement

Can a security training tool detect suspicious messages when attackers use unfamiliar wording, shortened links and harmless-looking urgency cues?

02 | Project Description

A helpdesk receives simulated emails and SMS. Training data overrepresents obvious phishing keywords, while evaluation includes new tactics and legitimate messages that mention passwords or payroll.

03 | Project Objectives

At the end of this project, the student team must demonstrate the following measurable outcomes:

FR-01. Deduplicate near-identical templates before splitting train and test.

FR-02. Compare rules, TF-IDF linear classifier and a third chosen model.

FR-03. Use campaign-separated holdout to test generalization to unseen templates.

FR-04. Add reject/abstain option and safe explanation of risk cues.

FR-05. Measure false-positive rate on legitimate urgent messages and robustness drift.

04 | Scope and Exclusions

Classify controlled training messages by risk and explain suspicious cues without opening links or collecting live user secrets. This is an educational triage tool.

Student Assignment Register

TEAM / STUDENT ID	MENTOR ASSIGNED DATA / VARIANT
_____	_____

ML-07 / DATA & REQUIREMENTS

Technical requirements

All functional requirements must be demonstrated with evidence, test logs, and a requirement-to-test mapping.

05 | Dataset & Data Sources

Minimum schema / fields: message_id, channel, subject_or_title, body_text, sanitized_url_features, sender_domain_type, label, campaign_group, collected_date. 400+ public/licensed or generated examples with provenance.

Allowed dataset source: Open legally licensed security awareness corpora and locally authored benign/phishing simulations; all URLs defanged and attachments excluded.

06 | Functional Requirements

ID	REQUIRED SYSTEM BEHAVIOUR
FR-01	Deduplicate near-identical templates before splitting train and test.
FR-02	Compare rules, TF-IDF linear classifier and a third chosen model.
FR-03	Use campaign-separated holdout to test generalization to unseen templates.
FR-04	Add reject/abstain option and safe explanation of risk cues.
FR-05	Measure false-positive rate on legitimate urgent messages and robustness drift.

07 | Non-Functional Requirements

ID	QUALITY / SAFETY REQUIREMENT
NFR-01	Do not click, fetch or execute suspicious content.
NFR-02	No real credentials, mailbox dumps or personal information allowed.
NFR-03	Clearly display that classification is decision support, not a guarantee of safety.

08 | Unique Team Constraint

Mentor assigns an unseen synthetic campaign plus a different legitimate business message set for evaluation.

The mentor will define a team-specific input variation and holdout sample. The final evaluation will include data or scenarios not supplied in an online tutorial.

ML-07 / IMPLEMENTATION PLAN

Student tasks & tests

09 | Required Student Tasks

TASK	WORK TO COMPLETE
T1 / Definition	Write a one-page use-case, target/output definition, decision user, exclusions and a requirement traceability table.
T2 / Data acquisition	Gather permitted data, record exact source/licensing, fix schema inconsistencies, and keep unchanged raw records.
T3 / EDA and validity	Document missing values, distributions, outliers, grouping, time ordering and target/feature leakage risks.
T4 / Baseline and modelling	Implement a simple baseline plus suitable ML alternatives, with a reproducible split and logged experiment settings.
T5 / Verification	Run the listed acceptance tests, inspect errors by data subset, and justify the metric and any prediction uncertainty.
T6 / Prototype	Build a small, reliable user interface or CLI with valid inputs/outputs, graceful error handling and saved model version.
T7 / Review-ready evidence	Maintain an experiment journal, repo history, source attribution, AI-usage disclosure, setup README and final report.

10 | Acceptance Test Cases

Students must test these specific conditions and attach screenshots/logs showing the actual outcome.

CASE	INSTRUCTOR / INPUT CONDITION	EXPECTED EVIDENCE
AT-01	Legitimate password-reset notice appears	Explain and report false-positive impact.
AT-02	Phishing uses unfamiliar phrasing	Assess on a campaign excluded from training.
AT-03	Defanged URL / punctuation pattern changes	Show shifts in probability and model limitation.

Submission Rule

Show the result of your own data partition and model. A copied notebook, generic dataset result, or UI screenshot without repeatable validation does not satisfy these tests.

ML-07 / CHECKPOINTS & REPORT

Milestones & final report

11 | Mentor Checkpoints

Each checkpoint requires proof and mentor feedback before proceeding. Changes must be documented in the experiment log.

REVIEW	CHECKPOINT / EVIDENCE TO PRESENT
CP-1	Inspect safe corpus handling and confirm all URLs are inert.
CP-2	Audit near-duplicate removal across splits.
CP-3	Require campaign-based test and confusion analysis, not random accuracy only.
CP-4	Review explanations for both a missed phishing message and flagged legitimate one.
CP-5	Present 8 new harmless simulated messages with controlled text perturbations.

12 | Final Report Format (Mandatory)

SECTION	WHAT TO INCLUDE
1. Cover & abstract	Project code/title, student/team IDs, mentor, 150-200 word abstract.
2. Problem & objectives	Local problem context, measurable outcomes, scope, exclusions, stakeholders.
3. Literature & sources	Minimum 3 relevant references plus dataset license/permission statement.
4. Data & EDA	Schema, cleaning log, missingness, statistics and 3 labelled visualizations.
5. Design & methods	System architecture, train/test strategy, baseline and model rationale.
6. Results & experiments	Metrics, baseline comparison, model table, 3 error examples, test logs.
7. Safety & limitations	Privacy, data suitability, failures, confidence and known constraints.
8. Conclusion & appendix	Practical findings, future improvements, source files, screenshots and repo link.

13 | Required Submission Artifacts

Corpus provenance sheet, sanitization log, deduplication proof, held-out campaign evaluation, risk-review UI, threat model.

Mandatory files: SRS.pdf, final_report.pdf, README.md, requirements.txt/environment.yml, source code, data/source manifest, results table, acceptance-test evidence and a 3-5 minute demonstration.

14 | Final Review / Verification

Student/team: _____ Mentor: _____ Date: _____

Checkpoint approval: CP-1 [ ] CP-2 [ ] CP-3 [ ] CP-4 [ ] CP-5 [ ] Revision required: Yes [ ] No [ ]

AI tools may assist learning, but the student must be able to reproduce all code, explain design choices, and complete a fair mentor-supplied live variation.